Unsupervised Building Change Detection in Remote Sensing Images via Joint Structural and Nonstructural Variation Generation

Yaoyao Wang, Jialei Yu, Han Wang, and Xinxiao Wu, *Member, IEEE*

***Abstract*—Remote sensing change detection (CD) plays a vital role in urban expansion analysis, ecological assessment, and disaster monitoring. While deep-learning-based methods have significantly advanced performance, they often rely on large-scale multitemporal datasets with pixel-level annotations. To alleviate this dependence, generating paired bitemporal remote sensing images have been explored, but its effectiveness is limited in the absence of pixel-level semantic masks. Moreover, most generation-based CD methods focus solely on modeling structural changes, such as the construction or removal of buildings while neglecting nonstructural variations (i.e., changes caused by illumination shifts, atmospheric fluctuations, or seasonal dynamics that do not alter the physical structure of objects). In this article, we propose SNV-GenCD, an unsupervised CD framework for buildings in high-resolution optical remote sensing images, which generates pseudobitemporal remote sensing images by jointly modeling structural changes and nonstructural variations. Within SNV-GenCD, a structural change generation module produces object-manipulated images along with corresponding pixel-level change masks, and a nonstructural variation module injects radiometric changes that reflect real-world remote sensing conditions. Furthermore, to eliminate temporal bias and enhance robustness against pseudochanges in the generated data, we introduce two complementary constraints that guide the detector to learn robust, order-invariant representations. Experiments on the LEVIR-CD, WHU-CD, and GoogleCD datasets show that SNV-GenCD achieves $F1$ scores of 82.2%, 87.6%, and 76.7%, respectively. These results significantly outperform existing state-of-the-art (SOTA) unsupervised methods and are competitive with weakly supervised approaches. The code is available at https://github.com/Shakeyoo/SNV-GenCD**

***Index Terms*—Change detection (CD), data generation, diffusion model, unsupervised learning.**

I. Introduction

REMOTE sensing change detection (CD) aims to identify object changes, by analyzing images of the same area acquired at different times, thus providing spatio-temporal information support for tasks such as Earth dynamics observation [1], urban expansion analysis [2], [3], disaster assessment [4], [5], and land-resource management [6]. With the increasing availability of high-resolution remote sensing data, deep-learning-based CD methods have become an indispensable tool for large-scale and timely environmental monitoring. However, building CD, as a representative and practically important branch of remote sensing CD, remains highly challenging due to complex urban backgrounds, diverse building scales and shapes, seasonal and radiometric changes, and the difficulty of obtaining high-quality, temporally aligned, pixel-level annotations.

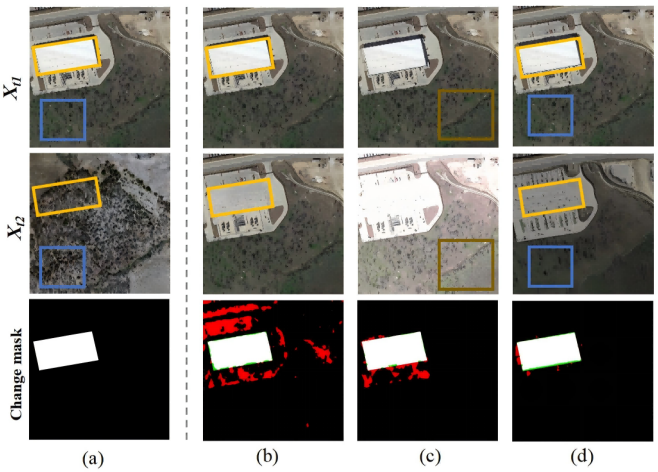


Fig. 1. Examples of different bitemporal image pairs. (a) Real bitemporal images, with blue and yellow rectangles indicating the pseudochange and structural change. (b) Pseudobitemporal images generated by inpainting showing only structural change. (c) Traditional environmental simulation using color balance, brightness, contrast adjustments, and sharpening operations. (d) Pseudobitemporal images generated by our method. The third row displays the corresponding change maps. In the third row, white areas denote ground-truth change regions, while red and green regions indicate false positives and false negatives, respectively.

Traditional CD methods [7], [8], [9] typically rely on paired bitemporal images with pixel-level change masks [Fig. 1(a)], requiring domain expert knowledge and considerable effort for image collection, preprocessing, and annotation. To ease this burden, researchers have turned to generating paired pseudobitemporal images by manipulating single-temporal images through rule-based operations, such as random pairing within each minibatch [10], intraimage and interimage patch exchange [11], inpainting [12], and copy–paste augmenta-

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tion [13]. However, these methods focus solely on structural manipulations, such as adding or removing objects within images [see the yellow rectangles in Fig. 1(b)], and often struggle to preserve spatial coherence or to simulate complex nonstructural variations, including radiometric changes (e.g., illumination shifts and atmospheric fluctuations) and seasonal dynamics that are difficult to capture with traditional environmental simulation methods [see the brown rectangles in Fig. 1(c)]. Consequently, models trained on such pseudopairs may learn to detect only structural changes, leading to increased false positives during CD, as illustrated in the third row of Fig. 1(b).

To overcome the limitations of rule-based methods, this article explores the use of diffusion-based generative models [14], [15], which offer superior stability and higher fidelity compared to GANs [16], to generate diverse and semantically coherent pseudobitemporal remote sensing data. These models are capable of generating high-fidelity images conditioned on semantic masks, thereby enabling the simulation of plausible scene manipulations. By editing a semantic mask to reflect a desired change and inputting it into a trained generator, a new image can be produced and paired with the original to form a pseudobitemporal sample. However, these generative models typically require large-scale annotated datasets and significant computational resources for effective training. In the absence of sufficient supervision, the generated outputs may exhibit structural inconsistencies and struggle to reproduce radiometric changes, which are essential for reliable CD.

In this article, we propose an unsupervised single-temporal remote sensing CD framework that jointly models both structural changes and nonstructural variations in remote sensing imagery. In particular, we design a structural change generation module that removes objects within a given image to simulate the desired object changes while preserving consistency in the surrounding appearance. Although structural manipulations can simulate object changes, it remains challenging to ensure robustness against pseudochanges caused by nonstructural disturbances. These variations are typically subtle, spatially continuous, and nonlinear, with most of their spectral effects concentrated in specific frequency bands. This makes them difficult to reproduce through rule-based methods or direct pixel-level augmentations, which generally lack frequency sensitivity. Given this challenge, frequency-domain manipulation offers a physically grounded and semantically consistent means of simulating radiometric changes, such as illumination shifts and atmospheric fluctuations. To this end, we additionally introduce a nonstructural variation module that simulates radiometric changes by applying controlled Fourier-domain perturbations in the latent feature space of a pretrained diffusion model.

Given that our generated bitemporal data exhibit a predefined change direction along with nonstructural variations, these characteristics present unique challenges during detector training and necessitate tailored constraints to ensure reliable change prediction. Based on these properties, we design two complementary constraints to guide the training of the change detector. First, we introduce a temporal symmetry constraint to promote prediction consistency regardless of the input order, under the assumption that the detected change should be invariant to temporal sequencing. Second, to enhance robustness against radiometric perturbations, we propose a perturbation consistency contrastive constraint that enforces stable predictions in unchanged regions, enabling the model to distinguish true changes from pseudochanges caused by illumination shifts, atmospheric fluctuations, and seasonal dynamics.

Our main contributions are summarized as follows.

1) We propose SNV-GenCD, an unsupervised CD framework for buildings in remote sensing images that jointly models structural changes and nonstructural variations to generate pseudotraining data and improve model generalization. By performing structural manipulation and radiometric perturbations on a single-temporal remote sensing image, the framework generates controllable pseudobitemporal pairs without relying on real temporal data or manual pixel-level annotations.
2) We propose a frequency-domain perturbation strategy to simulate nonstructural variations. Instead of retraining a generative model, our method perturbs the latent representation of a pretrained diffusion model in the Fourier domain, providing a low cost and physically grounded way to simulate radiometric shifts in complex remote sensing scenarios.
3) We propose two complementary constraints, a temporal symmetry constraint and a perturbation consistency contrastive constraint, to train our CD model. These constraints enhance the model's ability to distinguish changed from unchanged regions even in the absence of ground-truth change masks.

II. Related Work

A. Supervised and Semisupervised CD

Remote sensing CD has undergone significant evolution over the past decades, progressing from traditional feature processing approaches to deep-learning-based paradigm [17], [18], [19], [20]. Supervised methods rely on paired bitemporal images with pixel-level annotations. Early CNN-based architectures, such as FCNs [7] and densely connected Siamese networks [8], have laid the foundation by exploring feature fusion and shallow-feature exchange. More recent models, such as UNet3+ [21], enhance multiscale representation and improve change localization accuracy in high-resolution imagery. Subsequent works addressed issues like class imbalance and boundary precision through sampling strategies and attention mechanisms [22]. While transformers [23] enabled long-range dependence modeling and inspired token-based frameworks [24], Swin Transformer encoders [25], object-guided Transformer architectures [26], and hybrid CNN–Transformer designs [27], their quadratic complexity remains a computational bottleneck. Recently, state-space models (SSMs) [28] such as Visual Mamba [29] and Vision Mamba [30] have emerged as efficient alternatives, and their adaptation to CD (e.g., CDMamba [31], MambaCD [32])

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has achieved state-of-the-art (SOTA) results by integrating global–local interactions.

Semisupervised and weakly supervised CD methods aim to reduce the dependence on fully annotated bitemporal data by introducing limited or coarse supervision. Semisupervised approaches typically combine a small amount of labeled samples with a large set of unlabeled data. Early approaches focused on metric learning and semisupervised classifiers (e.g., S^3VM [33], Gaussian Processes [34]), while recent deep-learning methods [35] heavily exploit generative adversarial networks (GANs) to enhance pseudolabels or generate synthetic samples. In addition, consistency regularization [36] has emerged as a key trend, enforcing model robustness through input perturbations and contrastive learning strategies.

B. Unsupervised CD

In unsupervised CD, paired bitemporal images are provided without manually annotated masks. Existing methods typically obtain supervisory information from the intrinsic distributions or correlations of the input data, and then design models to learn discriminative change representations. Zhang et al. [37] employ a deep belief network to extract change features, while Wu et al. [38] introduce a kernel-PCA convolutional module for unsupervised spatio-spectral representation and build a corresponding deep network. Graph convolutional networks have also been applied to capture nonlocal spatial and temporal dependencies in remote sensing imagery. Although these methods eliminate the annotation burden, their lack of supervisory constraints weakens their feature representation capacity, limiting performance in deep architectures. Accordingly, they are often only applied to small regions or relatively simple change scenarios.

With the advent of vision foundational models, zero-shot CD leveraging large pretrained architectures has achieved breakthrough results in remote sensing. For instance, SCM [39] integrates the fine-grained segmentation capability of SAM with the cross-modal semantic understanding of CLIP [40], constructing a multiscale feature recalibration mechanism and a segmentwise semantic attention scheme to suppress spurious changes in an unsupervised setting. Similarly, segment any change [41] exploits spatiotemporal semantic similarity in SAM's latent space, performing bitemporal latent matching and point-query operations for zero-shot CD. However, these latent-space similarity assumptions may fail under significant illumination differences or when changes are very subtle, and the point-query mechanism requires manual specification of objects of interest, hindering fully automated processing of large-scale data.

Existing unsupervised methods are typically built on real bitemporal images and learn change representations directly from them. On the contrary, our method starts from single-temporal images and generates pseudobitemporal samples for model training.

C. Weakly Supervised CD

Weakly supervised methods rely on more economical supervision signals, including image-level, patch-level, or point-level labels, to guide model training [42], [43], [44]. Representative works include CS-WSCDNet [42], which refines image-level supervision using CAM and SAM, as well as transformer-based or generative frameworks that suppress spurious changes or unify different supervision regimes [43], [44]. Despite alleviating annotation costs, both semisupervised and weakly supervised methods often depend on auxiliary supervision or pseudolabels, whose quality can significantly affect robustness and generalization.

A related line of research investigates single-temporal CD, which can be regarded as a special form of weak supervision. Instead of using manually annotated bitemporal change masks, these methods construct pseudobitemporal training pairs from single-temporal images with segmentation masks. According to how pseudopairs are generated, existing methods can be roughly divided into rule-based construction methods and generative construction methods. Rule-based methods are defined as approaches that construct pseudochange pairs by applying predefined operations, such as permutation, patch exchange, or logical operations, to existing single-temporal images and their semantic masks. Zheng et al. [10] introduced the STAR method, which randomly permutes images within each training batch to form pseudobitemporal pairs and derives binary change masks from single-temporal binary object semantic masks via a logical XOR operation. This method enables high-precision CD without requiring any paired masks. Seo et al. [45] proposed self-pair, which swaps image patches within each image and selectively copies or removes semantic instances to produce semantically consistent pseudochange pairs. Chen et al. [11] extended this idea by exchanging patches both within and across images, thus increasing the diversity of generated samples and reducing overfitting during training.

Generative construction methods refer to approaches that leverage generative models to produce pseudobitemporal data directly from single-temporal images. Zheng et al. [46] further developed the Changen and subsequently extended it to Changen2 [47] frameworks. Changen formulates a generative probabilistic change model (GPCM) that decouples semantic-state transitions from image-level change generation into two subprocesses, allowing it to produce change sequences of arbitrary length and resolution from a single-temporal image. Changen2 formulates a GPCM with diffusion transformers to generate multitemporal image series and corresponding change masks from single-temporal data. Wang et al. [48] exploited a semantic-guidance diffusion model to generate pseudo-bitemporal pairs and introduced a refined loss to heighten sensitivity to changed regions, thus boosting generalization. To tackle the scarcity of building-change examples, Li et al. [49] presented a metric-guided single-temporal learning framework: it generates pseudobitemporal building samples via patch pairing and integrates a multigranularity spatial attention module (MGSAM) to mitigate geometric misalignment and radiometric changes, improving spatial-context modeling.

Despite their advantages, these methods still rely on high-quality semantic masks for single-temporal images, which are costly to obtain, and they struggle to account for the radiometric changes introduced by varying imaging conditions.

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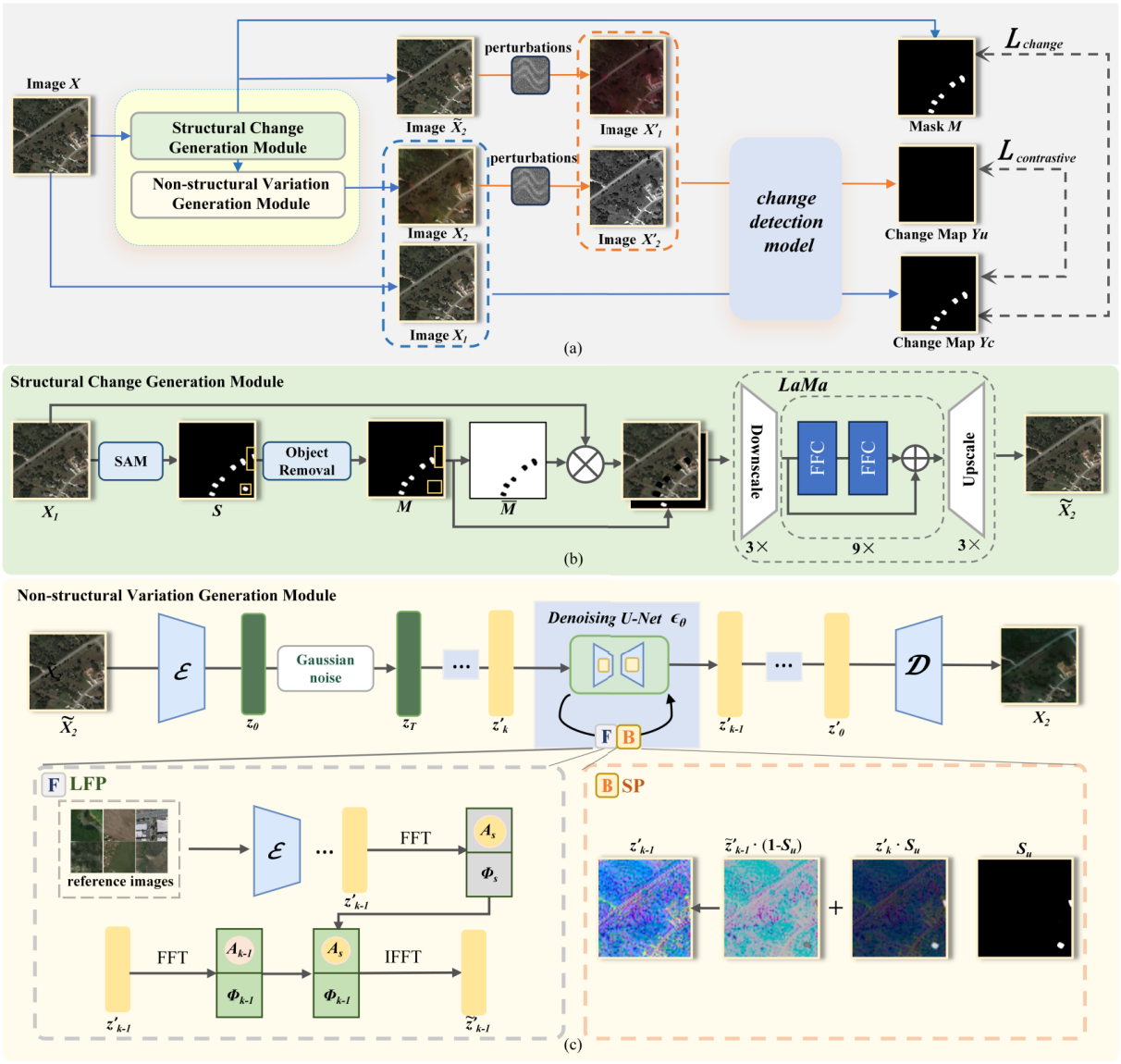


Fig. 2. (a) Overview of the proposed framework. The blue line denotes the main training path, while the orange line denotes the contrastive learning path. The gray dashed arrows represent the temporal symmetry constraint and the perturbation consistency contrastive constraint applied during training. (b) Structural change generation module. (c) Nonstructural variation generation module.

Environmental simulation techniques reported in [11] and [50] generally perform simple pixel-level image augmentations such as changes in brightness or contrast; however, these operations provide only coarse control over local details in high-resolution imagery and cannot faithfully reproduce the complex environmental variations encountered in real remote sensing scenes.

III. Method

A. Overview

Given two images captured at the same geographic location but at different times, denoted by $X_{t_1} \in \mathbb{R}^{H\times W\times C}, X_{t_2} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height, width, and number of channels, respectively, the goal of unsupervised CD is to train a binary change detector D without any paired supervision or labeled data. The detector outputs a change map $Y \in \{0,1\}^{H\times W}$, where 1 denotes changed pixels and 0 denotes unchanged ones, indicating changes between X_{t_1} and X_{t_2}.

In this article, we propose to generate pseudobitemporal pairs (X_1, X_2) from a single-temporal image X. In particular, we set $X_1 = X$, and X_2 is generated in two sequential stages. Structural manipulations (e.g., object removal) are applied to X_1 to construct changed regions, producing an intermediate image $\widetilde{X}_2$. Then, nonstructural variations (e.g., radiometric changes such as illumination shifts and atmospheric fluctuations, or seasonal dynamics) are injected into $\widetilde{X}_2$, yielding the final X_2. These generated image pairs, along with their generated change masks, are then used to train the detector D. Our framework consists of three key components: 1) a structural change generation module; 2) a nonstructural variation generation module; and 3) a change detector. The overall pipeline is illustrated in Fig. 2(a).

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The structural change generation module applies object manipulations to the input image X to generate object changes and produces a corresponding binary change mask $M \in \{0,1\}^{H\times W}$. The nonstructural variation module further manipulates the frequency domain to simulate realistic radiometric changes. The resulting pseudobitemporal image pairs are then used to train a change detector, guided by two constraints: temporal symmetry, which mitigates bias induced by temporal input order; and perturbation consistency, which enhances robustness to nonstructural variations. Thus, we can enable effective CD without requiring annotated bitemporal data. In the following sections, we provide detailed descriptions of each component in the proposed framework.

B. Structural Change Generation

The structural change generation module produces both pseudobitemporal pairs $(X_1, \widetilde{X}_2)$ and corresponding binary change masks by simulating spatial object changes, such as the objects' insertion or removal from the input image X. In this work, we simulate structural changes solely through "object removal," rather than object insertion, since generating entirely new objects often introduces unnatural artifacts [13] or *hallucinations* [51], [52] that may confuse the change detector. In particular, we treat the original input X as X_1, and generate the manipulated image $\widetilde{X}_2$ by removing objects using the generated pixel-level change masks. Fig. 2(b) displays our structural change generation process.

As the generated image pairs only simulate object removal, this asymmetry may bias the model toward one-sided changes. Therefore, we introduce a temporal symmetry constraint to alleviate the overfitting problem in binary object CD. The design and implementation of this constraint will be detailed in Section III-D.

We first adopt a segment anything model (SAM) [53] specifically designed for remote sensing building extraction, rather than the original interactive inference mode of SAM. This adapted model is capable of automatically generating coarse priors of building regions without requiring any interactive prompts. For each image, we directly apply SAM to generate a set of object regions, which are then merged into a binary semantic mask $S \in \{0,1\}^{H\times W}$, where 1 indicates object pixels and 0 denotes background. To simulate structural changes, we identify connected components in S as individual objects. A random subset of these objects is selected for removal, and their corresponding regions are set to 0 in S to obtain the binary change mask M, as illustrated in Fig. 2(b), which highlights the region of changes. We then generate a masked image by removing the corresponding regions from the original image using the inverse of the change mask $\overline{M}$. This masked image $X \odot \overline{M}$ is stacked with the mask M itself to form a four-channel input tensor

$$T = \text{stack}\left(X \odot \overline{M}, M\right). \tag{1}$$

We then feed T into the pretrained LaMa [12] inpainting model to reconstruct the object-removal regions with texture-consistent background content, thereby producing the structurally manipulated image $\widetilde{X}_2$.

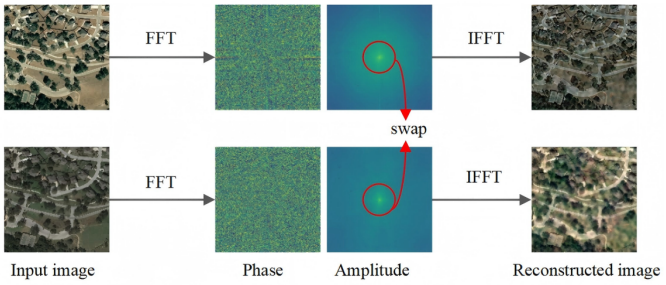


Fig. 3. Examples of low-frequency amplitude swapping. Two different remote sensing images are processed by FFT to produce their phase spectra and amplitude spectra, with the low-frequency regions of the amplitude spectra highlighted in red. The low-frequency amplitude components are then exchanged between the two amplitude spectra at corresponding positions, and an inverse FFT is applied to reconstruct the images. The reconstructed images demonstrate that swapping only the low-frequency amplitude components changes the global appearance while preserving edges and object structure.

C. Nonstructural Variation Generation

While the structural change generation module generates $\widetilde{X}_2$ to reflect object changes, it lacks the nonstructural variations typically observed in natural bitemporal remote sensing pairs. These nonstructural variations can introduce pseudochanges, potentially making the detector trained on such data vulnerable to false positives or poor generalization in real-world scenarios. In our framework, the Nonstructural Variation Generation module is implemented by two components: a low-frequency perturbation (LFP) block and a structure preservation (SP) block.

To simulate nonstructural variations while preserving structural consistency, we incorporate a frequency-domain perspective into the generation process, as illustrated in Fig. 2(c). According to Fourier analysis, an image's amplitude spectrum controls the distribution of energy across frequencies, thereby modulating global appearance (e.g., brightness and color), while the phase spectrum encodes local structural and spatial information [54]. Moreover, the amplitude spectrum can be further decomposed by frequency: its low-frequency components, concentrated in the center (as highlighted by the red circle shown in Fig. 3), represent smooth radiometric changes [55], whereas high-frequency components, located at the periphery, correspond to the strength of textures and object boundaries [56]. As demonstrated in Fig. 3, exchanging only the low-frequency amplitude components between two images while preserving the original phase allows us to manipulate the global appearance, without significantly altering spatial structure. We leverage this property in the latent space of Stable Diffusion by introducing an LFP block. This block selectively replaces the low-frequency amplitudes of the target latent representation with those from a reference image, enabling us to inject realistic radiometric changes while preserving semantic and spatial coherence. Fig. 1(d) shows one of the generated results.

To be more precise, given a structurally manipulated image $\widetilde{X}_2$, we first encode it into latent space using the VAE encoder $\varepsilon(\cdot)$ from a pretrained stable diffusion model [14]

$$z_0 = \varepsilon\left(\widetilde{X}_2\right). \tag{2}$$

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The latent representation z_0 is then forward diffused for T steps. Let $\{\beta_k\}_{k=1}^{T}$ be the noise schedule and define $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{i=1}^{k} \alpha_i$, resulting in a noisy latent representation z_k. At each step k, Gaussian noise is injected according to

$$z_k = \sqrt{\bar{\alpha}_k}\, z_0 + \sqrt{1-\bar{\alpha}_k}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{3}$$

The reverse diffusion process is then initiated from a noisy latent $z'_T = z_T$, where a pretrained denoiser ϵ_θ [14] predicts the noise component $\hat{\epsilon}$ at each step $k = T, \ldots, 1$, and the latent representation is progressively denoised

$$z'_{k-1} = \frac{1}{\sqrt{\alpha_k}} \left(z'_k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\, \hat{\epsilon} \right), \quad \hat{\epsilon} = \epsilon_\theta \left(z'_k, k \right). \tag{4}$$

At each intermediate step $k-1$, we apply a 2-D fast Fourier transform (FFT) to decompose z'_{k-1} into amplitude A_{k-1} and phase Φ_{k-1}

$$(A_{k-1}, \Phi_{k-1}) = FFT\left(z'_{k-1}\right). \tag{5}$$

To inject radiometric changes, we randomly sample a reference remote sensing image and obtain its amplitude spectrum A_s through the same encoding and an FFT process. We then define a central frequency radius $\rho = 1/2\sigma \cdot min(h, w)$ to isolate the low-frequency components of the amplitude spectrum, where σ is a relative frequency bandwidth ratio that controls the injection magnitude, and h and w denote the 2-D of the latent representation. Notably, σ regulates the extent of low-frequency components associated with large-scale radiometric variations. Within this region, we update the amplitudes as follows:

$$A_{k-1}(u, v) = \begin{cases} A_s(u, v), & \text{if } u^2 + v^2 \le \rho \\ A_{k-1}(u, v), & \text{otherwise.} \end{cases} \tag{6}$$

Here, u and v denote the frequency domain coordinates. According to (6), only the low-frequency components of the amplitude spectrum are replaced, while the high-frequency components remain unchanged. This ensures that fine-grained textures and object boundaries are preserved, preventing structural distortions. By expressing ρ as a function of $\min(h, w)$, this formulation maintains consistent LFP behavior across different image resolutions. We then reconstruct the latent representation z'_{k-1} with $\tilde{z}'_{k-1} = \text{IFFT}(A_{k-1}, \Phi_{k-1})$ to apply frequency-domain perturbations in the latent space, where $\text{IFFT}(\cdot, \cdot)$ is the inverse FFT.

To control the injection location of the perturbations, we propose a SP block, which performs a pixelwise replacement between the perturbed latent representation $\tilde{z}'_{k-1}$ and the original latent representation z'_k, guided by the unchanged region mask $S_u = S \oplus M$

$$z'_{k-1} = \tilde{z}'_{k-1} \odot (1 - S_u) + z'_k \odot S_u. \tag{7}$$

After completing the reverse diffusion steps, we decode the final latent representation z'_0 using the stable diffusion decoder to obtain the final image X_2, which, together with the original image $X_1 = X$ and the binary change mask M, forms a pixel-level labeled pseudo bitemporal training pair (X_1, X_2) for the change detector.

D. Training With Complementary Constraints

Since the structural change generation process involves only the removal of objects rather than insertion, the generated data exhibit a predefined change direction. In addition, non-structural variations introduced via LFPs enhance the visual diversity of the generated images at time t_2. Training on such data without proper regularization may lead the model to learn temporal bias or become overly sensitive to input order and simulated environmental noise. To effectively leverage these properties while mitigating potential biases, we train the CD model with two complementary constraints: a temporal symmetry constraint that prevents temporal bias by enforcing invariance to input order, and a perturbation consistency contrastive constraint that suppresses false positives in unchanged regions by enhancing robustness against simulated nonstructural variations. We implement these constraints on a Mamba-based CD model [32], which builds on the Visual Mamba [29] architecture and offers an efficient yet expressive design for modeling spatio-temporal dependencies in remote sensing imagery.

1) Temporal Symmetry Constraint: In binary object CD, the ground-truth change mask should remain consistent regardless of the temporal order of the input images. That is, object changes between X_{t_1} and X_{t_2} should satisfy the symmetry condition $Y_{t_1 \to t_2} = Y_{t_2 \to t_1}$. This temporal symmetry reflects the undirected nature of object changes, where the change itself is independent of image ordering [10]. However, due to our data generation process involving only unidirectional object removal, the temporal symmetry constraint does not require the generation process itself to be bidirectional or reversible. Instead, we explicitly train with both input orders (X_1, X_2) and (X_2, X_1) for each pair and impose the temporal symmetry constraint so that $Y_{t_1 \to t_2} = Y_{t_2 \to t_1}$. This strategy counteracts any potential bias from one-way removal and encourages the model to learn structural differences that are consistent for both object additions and removals.

In particular, for each generated pseudo bitemporal image pair X_1 and X_2, we feed both (X_1, X_2) and (X_2, X_1) into the detector and promote consistency between the two predicted change maps. The model is trained using a combination of binary cross-entropy loss ($\mathcal{L}_{ce}$) and Lovász-Softmax loss ($\mathcal{L}_{lov}$) [57], the latter of which is used to alleviate the imbalance between changed and unchanged pixels. Both losses are applied symmetrically to both input orders

$$\begin{aligned} \mathcal{L}_{change} = {} & \frac{1}{2} \left(\mathcal{L}_{ce}(X_1, X_2) + \mathcal{L}_{ce}(X_2, X_1) \right) \\ & + \mu \cdot \frac{1}{2} \left(\mathcal{L}_{lov}(X_1, X_2) + \mathcal{L}_{lov}(X_2, X_1) \right). \end{aligned} \tag{8}$$

Here μ is a balancing coefficient controlling the contribution of the Lovász-Softmax term. The binary cross-entropy loss is defined as

$$\mathcal{L}_{ce} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log p_i + (1 - y_i) \log (1 - p_i) \right] \tag{9}$$

where N is the total number of pixels in the change mask M, y_i is the ith pixel in M, and p_i is the model's predicted probability that the ith pixel has changed.

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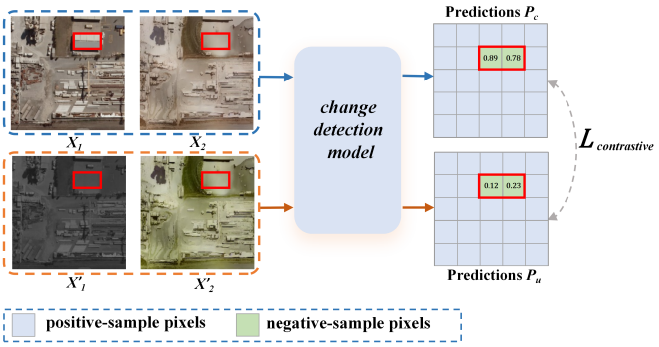


Fig. 4. Illustration of the perturbation consistency contrastive constraint. Blue arrows denote the main training flow, while orange arrows indicate the contrastive constraint training flow. The region inside the red box corresponds to the changed region, and the remaining area to the unchanged region.

2) Perturbation Consistency Contrastive Constraint: In addition to temporal symmetry, we aim to ensure robust prediction in unchanged regions even under varying imaging conditions. Bitemporal remote sensing images of the same location are often captured under different lighting, seasonal, or atmospheric conditions, leading to nonstructural variations that may result in pseudochanges. These pseudochanges may cause models to produce false positives. In this article, a perturbation consistency contrastive constraint is introduced to encourage the model to maintain prediction stability in unchanged areas while remaining sensitive to real structural changes.

As shown in Fig. 4, we generate an unchanged image pair (X'_1, X'_2) by applying random strong perturbations to X_2 and X_2, including colorjitter, grayscale, blur, and cutout. Since no object is added or removed between X'_1 and X'_2, they form an unchanged pair.

For the changed image pair (X_1, X_2), the CD model obtains their change probability map P_c. For the unchanged image pair (X'_1, X'_2), the predicted change probability map is P_u. Pixels inside the unchanged regions (blue areas in the probability maps of Fig. 4) constitute positive samples, where the model's predictions should consistently indicate unchanged, even with perturbations.

We define the positive sample loss $\mathcal{L}_p$ as the mean-squared error (mse) restricted to the unchanged region

$$\mathcal{L}_p = \mathrm{MSE}(P_c, P_u) \times (1 - M). \tag{10}$$

On the contrary, pixels inside changed regions (green areas in the probability maps of Fig. 4) act as negative samples, and the model's predictions should be opposite. We further design a negative loss that minimizes the discrepancy between the change probability of one image pair and the unchanged probability of the other, and vice versa, thereby enforcing opposite predictions between P_c and P_u in the changed regions. Formally

$$\mathcal{L}_n = [\mathrm{MSE}(P_c, 1 - P_u) + \mathrm{MSE}(1 - P_c, P_u)] \times M \tag{11}$$

where $1 - P_u$ represents the predicted unchanged probability for the unchanged pair (X'_1, X'_2), and $1 - P_c$ represents the predicted unchanged probability for the changed pair (X_1, X_2). Therefore, the contrastive loss is formulated as

$$\mathcal{L}_{contrastive} = \mathcal{L}_p + \mathcal{L}_n. \tag{12}$$

3) Total Loss: By combining the temporal symmetry constraint and the perturbation consistency contrastive constraint, we obtain the total loss

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{change} + \mathcal{L}_{contrastive}. \tag{13}$$

We integrate this total loss into the training process of the CD model. In particular, during each training iteration, the model is fed both original image pairs (X_1, X_2) and unchanged image pairs (X'_1, X'_2). We conduct ablation experiments to evaluate the impact of this loss by comparing model performance with and without it. The corresponding experimental results and discussion are presented in the following sections.

IV. Experiment

A. Datasets

We evaluate the proposed method on three building CD datasets in remote sensing images.

1) The LEVIR-CD [58] dataset comprises 637 high-resolution bitemporal image pairs from Google Earth, each sized 1024×1024 pixels at 0.5-m ground sampling distance. A total of 31 333 building-change instances are annotated, covering both building appearances and disappearances. The dataset is officially split into training, validation, and test sets in a 7:1:2 ratio. We use the images in the training set for data generation, and the test set for final performance evaluation.
2) The WHU-CD [59] dataset comprises two high-resolution aerial images of the same 20.5-km^2 area captured in 2012 and 2016. Each image measures $15\,354 \times 32\,507$ pixels at a ground sampling distance of 0.2 m, and contains 12 796 and 16 077 annotated building instances, respectively. WHU-CD is divided into training, validation, and test sets in an 8:1:1 ratio according to [59], and we follow the same data usage protocol as for LEVIR-CD.
3) GoogleGZ-CD [35] dataset is primarily focused on building CD, specifically capturing the dynamics of urban construction and demolition over a long time span. It comprises 19 remote sensing images from Guangzhou, acquired at different times between the years 2006 and 2019. These images exhibit substantial appearance differences caused by seasonal fluctuations. The image sizes range from $1,006 \times 1168$ pixels to 4936×5224 pixels and have the spatial resolution of 0.55 m per pixel. GoogleGZ-CD is processed following the same data usage protocol as WHU-CD.

B. Experimental Setup

1) Implementation Details: In our experiments, the proposed method is implemented in PyTorch and executed on a system with Python 3.9 and an NVIDIA GeForce RTX 4090 GPU (24 GB). We generate 3560 pairs of 512×512 images

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TABLE I
COMPARISON OF WEAKLY SUPERVISED AND UNSUPERVISED CD METHODS ON LEVIR-CD AND WHU-CD

Method	LEVIR-CD					WHU-CD				
	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)
• *Weakly Supervised*										
STAR [10]	97.6	63.5	77.7	73.5	82.3	98.1	64.5	78.4	76.4	80.5
Changen2 [47]	98.3	69.6	83.1	83.6	82.5	97.9	63.5	77.6	70.2	86.8
STCD [48]	94.7	21.5	35.4	36.7	34.3	98.3	64.7	78.6	79.2	78.3
MGF DNet [49]	98.1	61.8	76.4	81.5	71.9	98.9	76.8	86.9	89.3	84.6
B3-CDG [60]	97.4	59.6	74.7	74.3	75.0	97.7	58.5	73.8	71.6	76.1
• *Unsupervised*										
SCM [39]	89.5	19.1	32.1	23.9	48.7	86.3	15.2	26.4	17.0	59.2
OVCD [61]	96.5	53.5	69.7	62.8	78.2	97.8	61.1	75.8	71.3	80.9
CEG [62]	95.1	46.2	63.2	51.5	81.9	96.4	45.4	62.4	54.1	73.7
SNV-GenCD	**98.1**	**69.8**	**82.2**	**80.9**	**83.5**	**98.9**	**78.0**	**87.6**	**86.4**	**88.9**

from the LEVIR-CD dataset, 2,520 pairs of 512×512 images from the WHU-CD dataset, and 5764 pairs of 512×512 images from the GoogleGZ-CD dataset for training. Next, we train the CD model using the AdamW optimizer (learning rate 1×10^{-4}, weight decay 5×10^{-3}), a batch size of 2, and a total of 200 000 iterations. Unless otherwise specified, SNV-GenCD adopts MambaCD [32] as the default change detector.

2) Evaluation Metrics: We adopt four commonly used metrics to evaluate the performance of our CD model: precision (Prec.), recall (Rec.), $F1$ score, intersection over union (IoU), and overall accuracy (OA). The formulations are given as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

$$\text{F1} = \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}} \tag{16}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{17}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{18}$$

where TP is the number of true positive pixels (correctly predicted change), TN is the number of true negative pixels (correctly predicted unchanged), FP is the number of false positive pixels (incorrectly predicted change), and FN is the number of false negative pixels.

C. Results

1) Comparison With SOTA CD Methods: We compare our method with several SOTA unsupervised CD methods and weakly supervised single-temporal CD methods to validate the effectiveness of our method. The unsupervised methods include SCM [39], OVCD [61], CEG [62], while the weakly supervised single-temporal methods comprise STAR [10], Changen2 [47], STCD [48], B3-CDG [60], and MGFDNet [49]. The quantitative comparison results on the LEVIR-CD and WHU-CD datasets are reported in Table I, and those on the GoogleGZ-CD dataset are presented in Table II.

Table I shows that our method outperforms existing unsupervised methods. The high OA supports the $F1$ score improvements by suggesting the model effectively distinguishes unchanged regions, thereby minimizing false positives. Consistent performance gains can also be observed on the GoogleGZ-CD dataset in Table II, further demonstrating the robustness of our method under different data distributions.

TABLE II
COMPARISON OF WEAKLY SUPERVISED AND UNSUPERVISED CD METHODS ON GOOGLEGZ-CD

Method	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)
• *Weakly Supervised*					
STAR [10]	95.4	56.7	72.4	76.8	68.4
Changen2 [47]	94.8	54.1	70.2	71.4	69.1
STCD [48]	93.0	37.2	54.2	65.0	41.3
• *Unsupervised*					
SCM [39]	79.2	21.1	34.8	23.9	63.7
OVCD [61]	92.2	40.9	58.1	54.8	61.8
CEG [62]	89.9	35.4	52.2	44.8	62.7
SNV-GenCD	**95.7**	**62.1**	**76.7**	**72.9**	**80.9**

Moreover, our method achieves competitive results among weakly supervised methods, ranking first on most metrics across all three datasets. In particular, on the LEVIR-CD dataset, the $F1$ score of our method is slightly lower than that of Changen2. The advantage of Changen2 is attributed to its use of a diffusion model trained on a large amount of remote sensing data and semantic masks, enabling the generation of highly realistic pseudobitemporal image pairs that capture fine-grained structural changes. Similarly, on the WHU-CD dataset, MGFDNet slightly outperforms our method in precision, owing to its design of a metric-guided spatial attention module. In addition, on the GoogleGZ-CD dataset, our method achieves the best overall performance among all the compared methods, including weakly supervised ones, with a clear margin in IoU and $F1$ score. The improvement in OA is particularly meaningful given the substantial seasonal variations in GoogleGZ-CD that typically induce false positives. Although our precision is slightly lower than that of the STAR method, our $F1$ score outperforms STAR in this integrated evaluation. In particular, while STAR attains a higher precision than SNV-GenCD, its recall is markedly lower than ours, indicating that STAR tends to miss a considerable portion of true changed regions. On the contrary, SNV-GenCD

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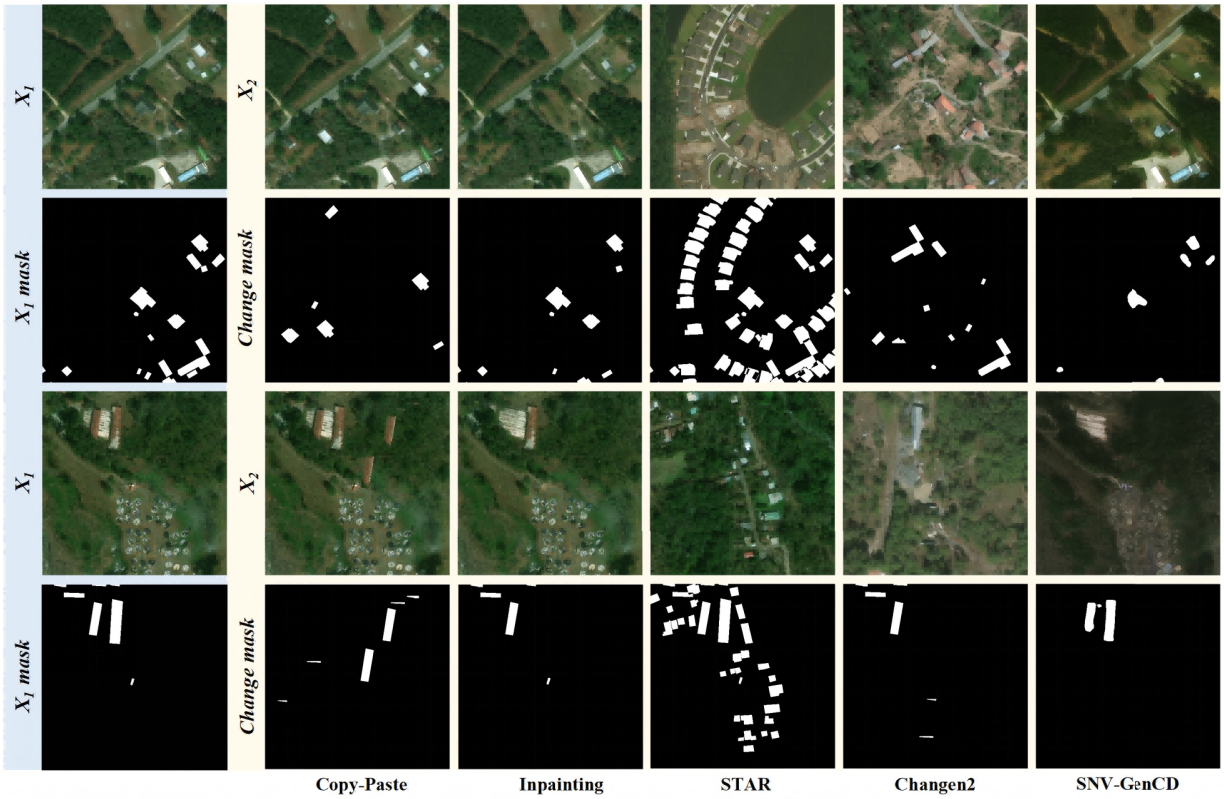


Fig. 5. Qualitative comparison of pseudobitemporal image generation methods. Each column corresponds to a specific generation method, and each row corresponds to a group of generated samples. The odd-numbered rows show the input single-temporal image X_1 and the generated image X_2, while the even-numbered rows show the semantic mask of X_1 and the corresponding change mask produced by each method.

recovers more real changes with only a moderate increase in false positives, which leads to a clearly higher $F1$ score and better overall detection performance. Despite these cases, our method still demonstrates strong generalization and robustness across datasets, confirming the effectiveness of our generative framework and complementary constraints even in comparison to weakly supervised frameworks.

2) Bitemporal Data Generation: In this subsection, we first provide a qualitative comparison of different pseudobitemporal data generation methods, as shown in Fig. 5. For each method, we visualize the real single-temporal input image X_1, the corresponding generated change mask, and the synthesized second temporal image X_2, enabling a direct comparison of their generation characteristics. Copy–paste and inpainting only modify local regions while keeping identical backgrounds outside the changed areas, resulting in limited appearance diversity. STAR constructs bitemporal pairs through random image pairing and XOR-based mask derivation, which often leads to spatial misalignment between X_1 and X_2. Changen2 generates more realistic structures using diffusion models, but may introduce geometric distortions in synthesized buildings. On the contrary, our method produces structurally coherent changes while simultaneously preserving diverse environmental variations across the scene. The generated X_2 not only exhibits object changes with smooth boundary transitions but also reflects global radiometric differences such as illumination and seasonal shifts, which are absent in the other methods.

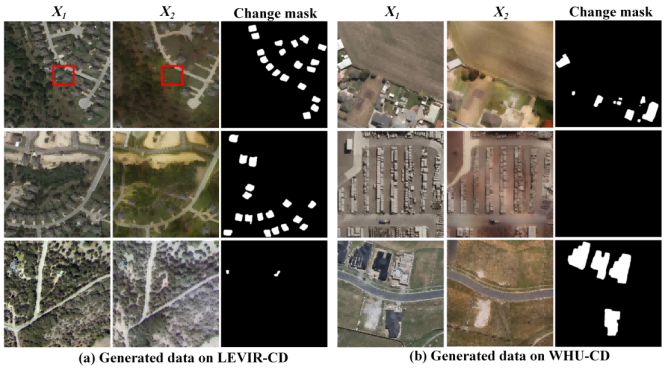


Fig. 6. Visualization of generated bitemporal image pairs. (a) Results on the LEVIR-CD dataset. (b) Results on the WHU-CD dataset. Each sample consists of three columns representing the single-temporal input image X_1, the generated image X_2, and the corresponding binary change mask from left to right.

We then focus on the qualitative results of our method on specific datasets. Fig. 6(a) and (b) display the results produced by our method on the LEVIR-CD and WHU-CD datasets, respectively. These results demonstrate that our method can effectively simulate environmental variations while enabling structurally coherent building removal and smooth integration with the surrounding area [as highlighted by the red rectangle shown in Fig. 6(a)]. In particular, the generated images reflect typical illumination changes and seasonal vegetation evolution, as illustrated by the first-row image pair in Fig. 6(a). In

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TABLE III
COMPARISON OF CD PERFORMANCE UNDER DIFFERENT PSEUDOBITEMPORAL DATA GENERATION METHODS (USING THE SAME CHANGE DETECTOR)

Method	LEVIR-CD					WHU-CD				
	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)
Copy-Paste [13]	83.1	11.5	20.7	13.6	43.4	73.3	12.4	21.0	12.0	85.9
Inpainting [12]	95.7	21.7	35.7	76.8	23.3	98.5	69.0	81.6	81.8	81.5
STAR [10]	97.5	63.0	77.3	72.0	83.4	97.9	63.3	77.5	71.0	85.3
Changen2 [47]	97.9	67.8	80.8	74.9	87.7	98.3	69.1	81.7	78.6	85.0
SNV-GenCD	**98.1**	**69.8**	**82.2**	**80.9**	**83.5**	**98.9**	**78.0**	**87.6**	**86.4**	**88.9**

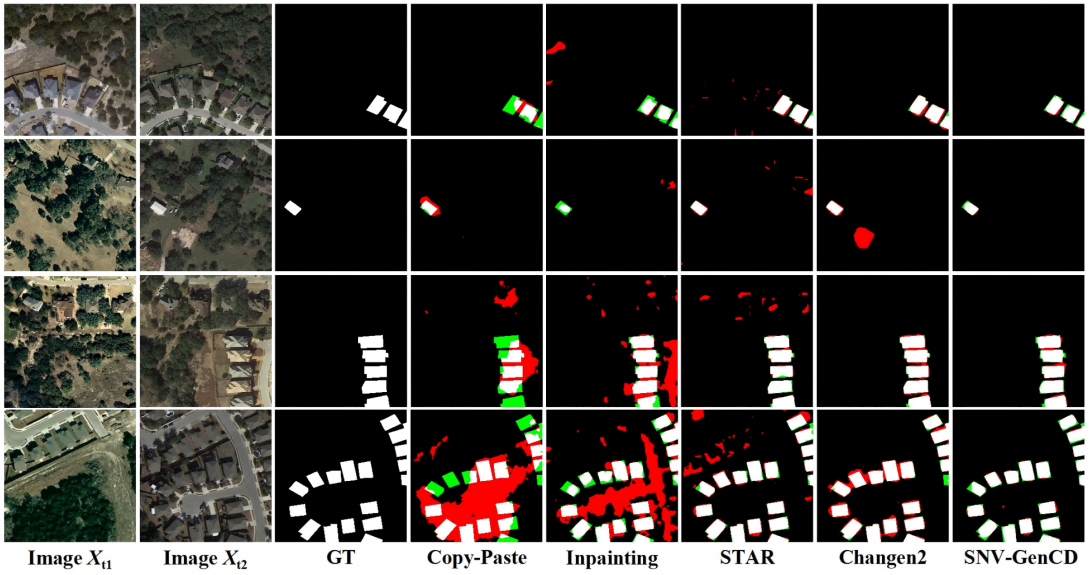


Fig. 7. Examples of change maps obtained by different methods on the test set of the LEVIR-CD dataset. In the obtained change maps, white represents TP, black represents TN, red represents FP, and green represents FN. Zoomed-in view for a better visual effect.

building removal scenarios, background consistency and smooth boundary transitions are maintained, and the surface textures and tones in the removed regions remain harmonious with adjacent areas without visible artifacts.

To further quantitatively assess how different pseudosingle-temporal generation strategies affect final detection performance, we evaluate our data generation method by using pseudobitemporal images generated by different methods as training data while keeping the same CD network architecture for fair comparison. We compare our method with four representative single-temporal generation methods: copy–paste [13], inpainting [12], STAR [10], and Changen2 [47]. Table III presents the quantitative results. Compared to the four methods, which rely on manually annotated semantic masks to generate pseudosamples, our method outperforms them on most evaluation metrics. Our generated images exhibit both realistic object changes and environmental variations. Based on their characteristics, we apply the temporal symmetry and the perturbation consistency as a complementary constraint strategy to help the model distinguish real changes from irrelevant perturbations, leading to high recall and improved precision.

Figs. 7 and 8 present change maps generated by our method alongside the four methods mentioned above on the LEVIR-CD and WHU-CD datasets. Copy–paste and inpainting only simulate structured object changes, resulting in maps with many false positives and false negatives. STAR produces bitemporal pairs through random pairing, but the results lack realism and still suffer from false alarms. Changen2 uses diffusion models to create more realistic bitemporal pairs, which reduces false positives and yields more complete change regions, yet it lacks explicit simulation of environmental variations. For example, in the second row of Fig. 7, a patch of land in the image at time t_2 is misclassified as a building change, and the textures generated by Changen2 sometimes do not match the true scene, leading to additional false detections. By contrast, our method has almost no false positives or false negatives, demonstrating that the generated pseudobitemporal pairs effectively enable CD models to learn realistic and discriminative change features.

3) Evaluate on Different CD Models: To evaluate the effectiveness of the generated pseudobitemporal data across different CD models, we conduct a comparative study under two training regimes: training on real annotated bitemporal data and training on pseudobitemporal pairs generated by our method. Evaluation is conducted on the real bitemporal test sets of LEVIR-CD and WHU-CD. The compared models are FC-EF [7], FC-Siam-Diff [7], SNUNet [8], ChangeFormer [63], Changer [64], TTP [65], ChangeStar [10], MambaCD [32], and our SNV-GenCD.

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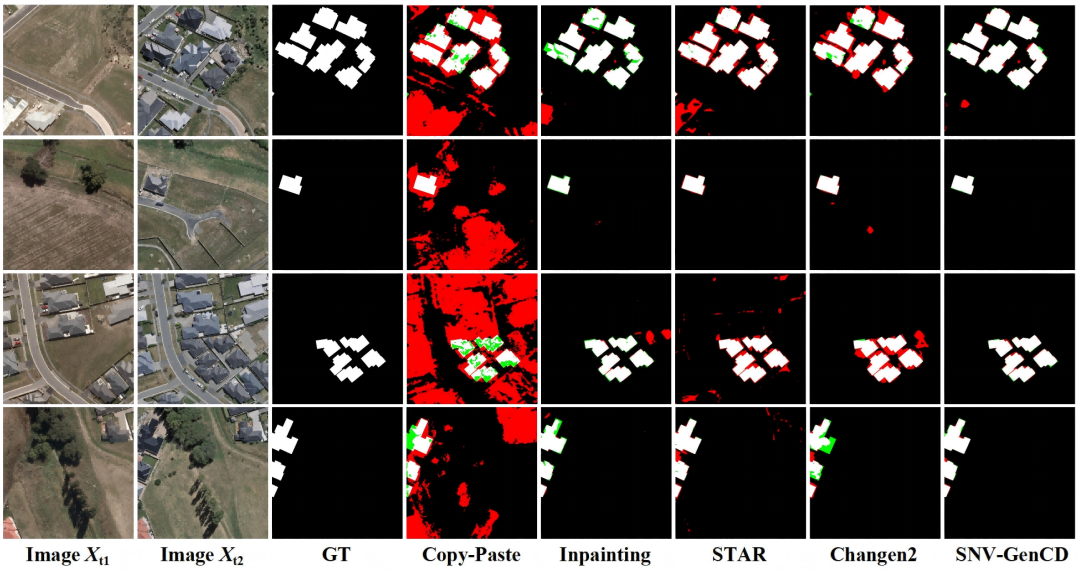


Fig. 8. Examples of change maps obtained by different methods on the test set of the WHU-CD dataset. In the obtained change maps, white represents TP, black represents TN, red represents FP, and green represents FN. Zoomed-in view for a better visual effect.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT CD MODELS TRAINED UNDER FULLY-SUPERVISED DATASETS AND OUR GENERATED DATA SETTINGS ON LEVIR-CD AND WHU-CD

Method	LEVIR-CD					WHU-CD				
	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)	OA(%)	IoU(%)	F1(%)	Prec.(%)	Rec.(%)
Fully-supervised (Real Bi-temporal Data)										
FC-EF [7]	98.9	75.2	85.8	87.4	84.2	98.8	73.7	84.8	83.5	86.3
FC-Siam-Diff [7]	99.0	77.4	87.3	91.1	83.7	99.1	78.0	87.6	90.8	84.6
SNUNet [8]	98.8	79.5	88.5	89.7	87.4	99.1	77.2	87.1	84.7	89.7
ChangeFormer [63]	98.9	79.9	88.8	90.6	87.0	99.2	82.3	90.3	93.7	87.1
Changer [64]	99.1	84.6	91.7	91.4	91.9	99.3	83.9	91.2	91.8	90.6
TTP [65]	99.1	85.1	91.9	92.1	91.7	99.2	85.3	92.0	92.1	91.9
ChangeStar [10]	99.0	82.8	90.6	92.0	89.2	99.3	84.2	91.4	90.7	92.2
MambaCD [32]	99.1	83.6	91.1	91.8	90.3	99.5	88.1	93.7	92.3	95.1
Single-temporal (Our Generated Data)										
FC-EF [7]	96.5	45.6	62.7	71.2	55.9	97.7	59.4	74.5	71.7	77.6
FC-Siam-Diff [7]	96.3	47.4	64.3	65.1	63.6	97.9	63.3	77.5	71.0	85.3
SNUNet [8]	97.2	59.4	74.5	71.7	77.6	98.1	64.5	78.4	76.4	80.5
ChangeFormer [63]	97.1	60.7	75.5	71.3	80.0	98.3	68.7	81.5	78.4	84.7
Changer [64]	97.4	61.9	76.5	72.6	80.7	98.4	69.0	81.6	81.8	81.5
TTP [65]	97.4	63.8	77.9	71.5	85.6	98.3	68.4	81.2	78.0	84.6
ChangeStar [10]	97.8	65.2	78.9	81.4	76.6	98.7	72.7	84.2	86.7	82.0
SNV-GenCD	**98.1**	**69.8**	**82.2**	**80.9**	**83.5**	**98.9**	**78.0**	**87.6**	**86.4**	**88.9**

Table IV summarizes the quantitative results on the two benchmark datasets. Overall, training on real annotated data generally yields stronger performance, as expected. However, when trained on the generated pseudobitemporal data, all evaluated models maintain reasonable performance, indicating that the proposed data generation strategy provides effective supervisory signals for CD. More advanced architectures, such as ChangeFormer, TTP, ChangeStar, and MambaCD, demonstrate smaller performance gaps between real-data supervision and pseudosupervision, suggesting that the modern detection models can better exploit the structural and radiometric cues embedded in the generated pseudodata.

In particular, our method, SNV-GenCD, achieves competitive results compared to its fully supervised counterpart, highlighting the potential of the proposed pseudodata as an alternative training source when manual annotations are limited or unavailable.

When applying our generated pseudobitemporal data to different CD models, we observe generally stable performance, though the best results are still achieved within our proposed framework. This suggests that while the generated data is broadly applicable, it performs most effectively when paired with our complementary constraints training strategy.

D. Sensitivity Analysis

We analyze the sensitivity of SNV-GenCD to two key factors: 1) the frequency bandwidth parameter σ in the LFP

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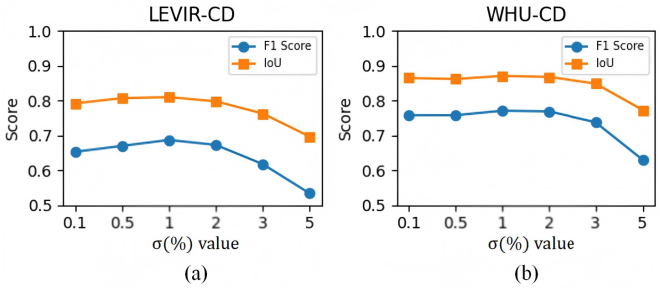


Fig. 9. Effect of the scale parameter σ on change-detector performance for two datasets. (a) LEVIR-CD dataset. (b) WHU-CD dataset.

TABLE V
PERFORMANCE UNDER DIFFERENT SCALES OF PSEUDOSAMPLE GENERATION ON LEVIR-CD AND WHU-CD

Pseudo Pairs	LEVIR-CD		WHU-CD	
	IoU (%)	F1 (%)	IoU (%)	F1 (%)
100	63.1	77.4	69.4	81.9
400	67.3	80.5	71.3	83.2
800	68.6	81.4	74.3	85.3

block and 2) the number of generated pseudobitemporal pairs used for training.

1) Sensitivity of σ: The scale factor σ is a critical hyperparameter in our LFP block, controlling the radius of low-frequency amplitude replacement. Fig. 9 evaluates model performance on LEVIR-CD and WHU-CD datasets in terms of $F1$ scores and IoU for $\sigma \in \{0.001, 0.005, 0.01, 0.02, 0.03, 0.05\}$. When σ increases from 0.001 to 0.01, $F1$ scores on the LEVIR-CD dataset rise from 0.792 to 0.81, and IoU from 0.653 to 0.687, while on the WHU-CD dataset $F1$ scores increase from 0.865 to 0.871 and IoU from 0.758 to 0.771. This indicates that a moderate expansion of the replacement radius enriches radiometric perturbations and thus enhances the detector's robustness to real changes. Beyond $\sigma = 0.01$, performance on both datasets declines: at $\sigma = 0.05$, the $F1$ score and IoU on the LEVIR-CD dataset fall to 0.697 and 0.534, and on the WHU-CD dataset to 0.773 and 0.63. This deterioration occurs because excessively low-frequency replacement produces overly smooth areas in the generated images; in these areas, object boundaries become blurred and unnatural artifacts appear, which weakens the model's ability to discriminate genuine changes. These results confirm that $\sigma = 0.01$ is the optimal setting for the LFP block in our method, significantly boosting unsupervised CD performance.

2) Sensitivity to the Number of Pseudopairs: To evaluate the practicality of our method in real-world applications where data collection is often costly or limited, we further investigated the model's performance with limited pseudoimage pairs. In particular, we simulated a few-shot generation scenario by restricting the number of generated pseudopairs with a resolution of 512×512 to varying scales: 100, 400, and 800.

Table V presents the quantitative results on the LEVIR-CD and WHU-CD datasets. The results demonstrate that our method maintains reasonable performance even with limited pseudoimage pairs. Notably, with only 100 pseudopairs, the model achieves an $F1$ score of 77.4% on LEVIR-CD and 81.9% on WHU-CD, indicating that the high-quality pseudosamples effectively capture the essential feature representations of change. Furthermore, as the scale of pseudodata increases from 100 to 800, we observe a consistent improvement in detection accuracy (e.g., $F1$ score improves by 4.0% on LEVIR-CD). This improvement mainly comes from the increased diversity of pseudopairs, which enriches the change patterns seen during training. The smaller gain from 400 to 800 pairs indicates a gradual saturation trend, suggesting that even a limited number of high-quality pseudopairs can already support effective representation learning.

TABLE VI
ABLATION STUDY ON LEVIR-CD AND WHU-CD

Step Configuration				LEVIR-CD		WHU-CD	
LFP	SP	Symmetry	Con	IoU(%)	F1(%)	IoU(%)	F1(%)
				21.7	35.7	65.3	79.0
		✓	✓	22.6	36.8	69.4	81.9
✓				59.9	74.9	71.2	83.2
✓	✓			65.6	79.2	73.7	84.8
	✓	✓	✓	66.7	80.0	73.1	84.5
✓		✓	✓	66.4	79.8	74.1	85.1
✓	✓		✓	68.1	81.0	75.9	86.3
✓	✓	✓		67.7	80.8	77.2	87.1
✓	✓	✓	✓	**69.8**	**82.2**	**78.0**	**87.6**

E. Ablation Study

Table VI presents the impact of each design element on overall performance. On both the LEVIR-CD and WHU-CD datasets, we conduct ablation studies on the LFP block, the SP block, the temporal symmetry constraint (Symmetry), and the perturbation consistency contrastive constraint (Con). In the table, a checkmark (✓) indicates that the element is enabled for that run, whereas a blank entry means it is removed.

Removing any one or a combination of these modules leads to a noticeable drop in performance. This demonstrates that each component contributes complementary strengths. In particular, the LFP block adds natural radiometric changes, the SP preserves spatial coherence, the temporal symmetry constraint alleviates the overfitting problem, and the perturbation consistency contrastive constraint enhances discrimination between changed and unchanged regions. These modules complement each other and together bring clear improvements across evaluation metrics.

The baseline only models structural changes, leaving unchanged regions strictly identical in the generated pseudopairs. WHU-CD consists of aerial imagery with relatively similar imaging conditions, where unchanged regions exhibit limited appearance variations. On the contrary, LEVIR-CD images from Google Earth have larger temporal gaps, resulting in significant seasonal and illumination variations even in unchanged regions. Since the baseline is trained only on pseudopairs with identical unchanged regions, it misclassifies these normal appearance variations as actual changes during testing, leading to severe false positives. This result demonstrates the importance of the LFP and SP modules in

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TABLE VII
COMPARISON OF NONSTRUCTURAL VARIATION GENERATION STRATEGIES ON THE LEVIR-CD AND WHU-CD DATASETS

Method	LEVIR-CD		WHU-CD	
	IoU (%)	F1 (%)	IoU (%)	F1 (%)
Image-level perturbation	50.6	67.2	70.2	82.5
Latent-space perturbation	**69.8**	**82.2**	**78.0**	**87.6**

TABLE VIII
COMPARISON OF MODEL COMPLEXITY ON THE LEVIR-CD DATASET

Method	Params(M)↓	GFLOPs(G)↓	FPS	IoU(%)
FC-EF [7]	1.35	3.24	92.28	45.6
FC-Siam-Diff [7]	1.35	4.38	64.65	47.4
SNUNet [8]	12.03	46.69	7.11	59.4
Changer [64]	47.48	29.46	19.98	61.9
ChangeFormer [63]	13.94	5.82	40.13	60.7
ChangeStar [10]	16.96	19.21	32.77	65.2
TTP [65]	353.26	292.17	1.50	63.8
SNV-GenCD	**49.94**	**57.41**	**39.20**	**69.8**

modeling nonstructural variations and enhancing robustness to appearance inconsistency.

To validate the effectiveness of our latent-space perturbation, we apply the same LFP directly to the raw images. As illustrated in Table VII, an LFP in the raw image yields significantly lower performance on both datasets compared to our latent-space perturbation method. Directly perturbing raw images merely alters global illumination and color, leaving land-cover textures unchanged, which fails to capture the holistic appearance shifts typical of real bitemporal pairs. On the contrary, the latent space of Stable Diffusion jointly encodes the contextual relationships among scene elements, enabling frequency perturbations to propagate coherently across the entire scene and produce globally consistent radiometric variations, generating training samples that significantly enhance the generalization ability of the CD model.

F. Time Complexity Analysis

To evaluate the computational efficiency of SNV-GenCD, we compare its model complexity with other CD methods in terms of the number of parameters, GFLOPs, and FPS (frames per second), as reported in Table VIII. Compared with existing methods, SNV-GenCD maintains a moderate model complexity without introducing excessive computational overhead. Together with its superior detection performance, this result indicates that the proposed method achieves a favorable balance between efficiency and effectiveness, which supports its practical applicability.

V. CONCLUSION

In this article, we propose an unsupervised remote sensing CD framework, which jointly generates structural changes and nonstructural variations from single-temporal images. A structural change generation module simulates object change manipulations with corresponding change masks, while a nonstructural variation module introduces realistic radiometric differences by applying frequency-domain perturbations in the latent feature space of a pretrained diffusion model. These two controllable modules enable the construction of diverse and semantically coherent pseudobitemporal image pairs without relying on real temporal sequences or manual annotations. Leveraging the characteristics of the generated data, we further introduce two complementary training constraints: a temporal symmetry loss to reduce temporal bias, and a perturbation consistency contrastive loss to enhance robustness against pseudochanges caused by radiometric variations. Experimental results on two publicly available datasets demonstrate that our method facilitates the effective training of change detectors in a fully unsupervised setting and achieves superior performance across multiple benchmarks.

However, the structural change generation module relies on SAM-based foreground segmentation, making it primarily suited for object CD with well-defined boundaries (e.g., buildings), and its generalization to land use and land cover CD remains limited. In future work, we plan to extend our framework to semantic CD and explore lightweight generation pipelines to improve computational efficiency and broader applicability in complex remote sensing scenarios.

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